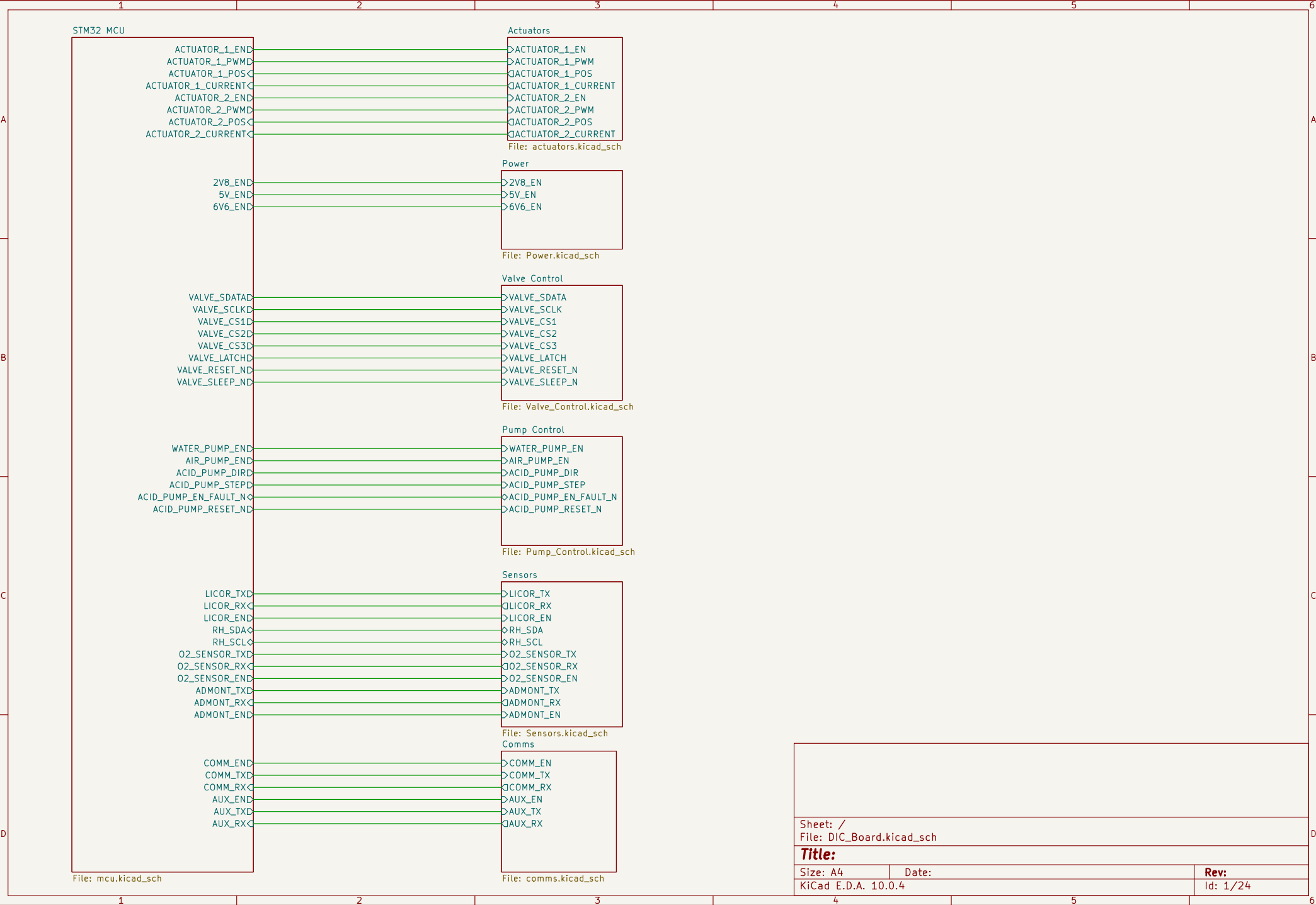
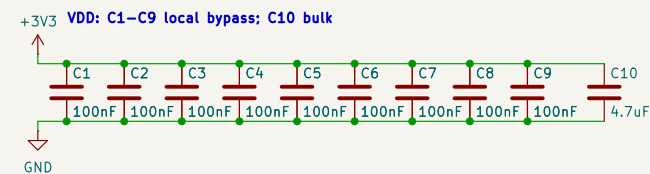


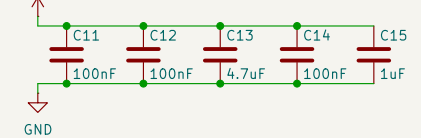


Sheet: /		
File: DIC_Board.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 1/24

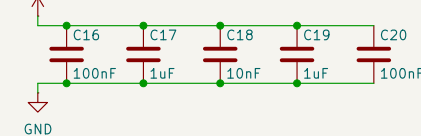
DECOUPLING CAPACITORS



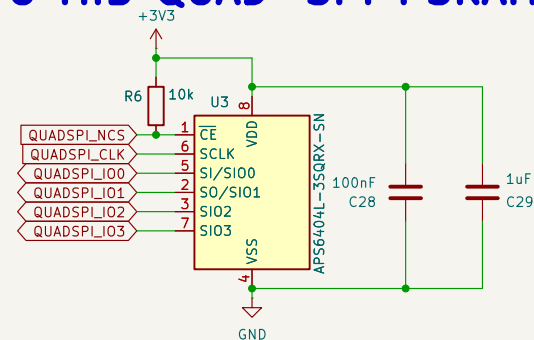
+3V3 VDDIO2: C11–C13; VDDUSB: C14–C15



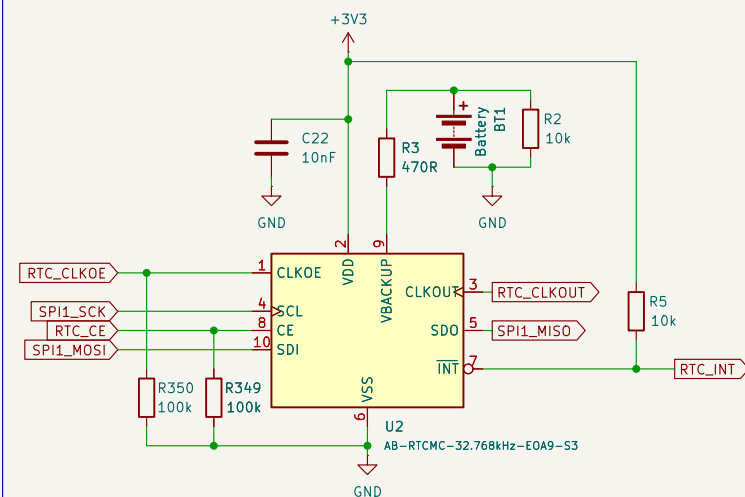
+3V3 VDDA: C16–C17; 3V3 analog bulk: C18–C19; VBAT: C20



8 MiB QUAD–SPI PSRAM



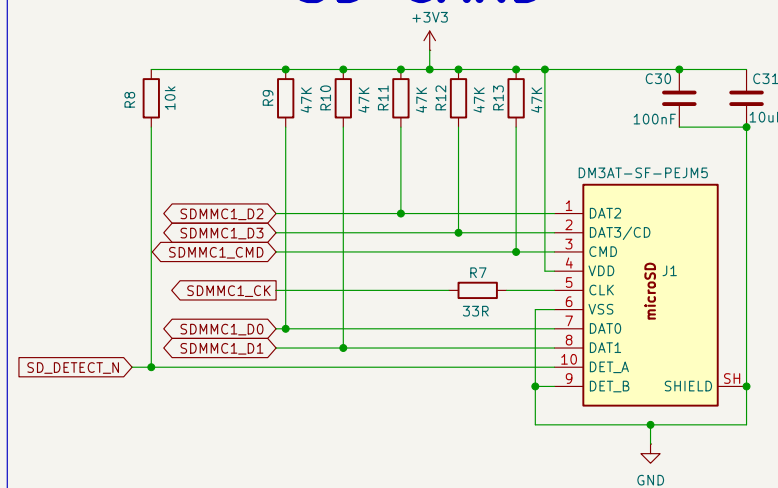
RTC



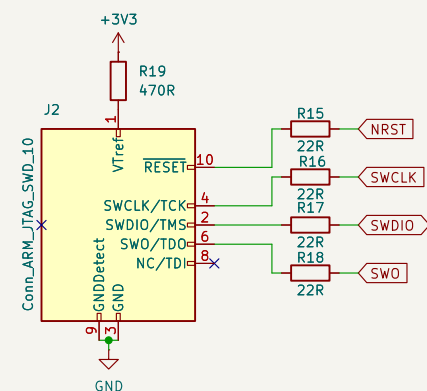
LSE qualification: C25/C26=6.8pF gives 3.4pF series load.
ABS06-107 requires effective 4.0pF, <=4.5pF including pin/PCB parasitics.
Tune these capacitors after layout and verify startup, gain and drive over temperature.
Do not treat the present values as final oscillator qualification.

VREF+: C27 100nF + C43 1uF must be local to U1 pin32, after FB1.
Keep VREFBUF output disabled when the external 3V3 reference feed is used.

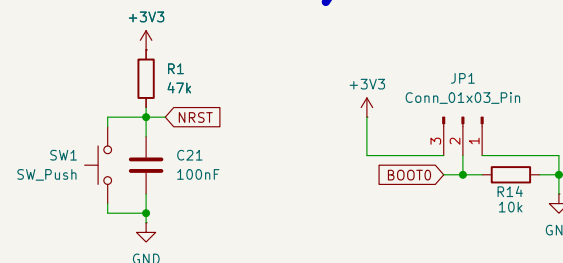
SD CARD



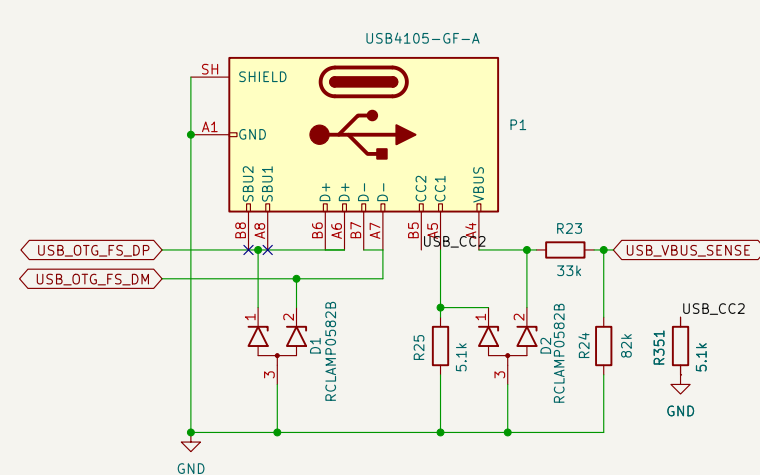
DEBUG



RESET/BOOT

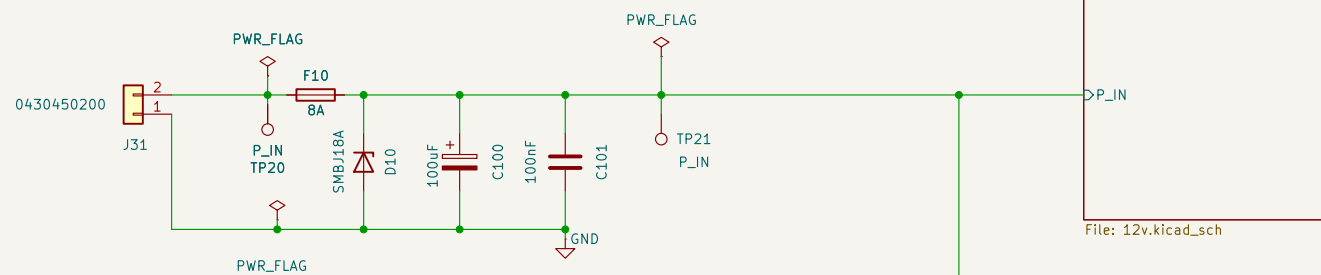


USB



Provisional 9–16 V operating envelope

Provisional 9–16 V operating envelope



12V

→ P_IN

File: 12v.kicad_sch

3V3

The diagram shows a 4-bit shift register. It has three inputs: **P_IN** (Parallel Input), **C_IN** (Clock Input), and **C_OUT** (Carry Out). It has four outputs: **Q₀**, **Q₁**, **Q₂**, and **Q₃**. The register is represented by a horizontal bar with four segments, each corresponding to one of the output bits. The inputs are connected to the register: **P_IN** to the first segment, **C_IN** to the clock input, and **C_OUT** to the carry out. The outputs **Q₀**, **Q₁**, **Q₂**, and **Q₃** are connected to the four segments of the register.

File: 3v3.kicad_sch

5V

→ 5V_EN

File: 5v.kicad_sch

6V6

6V6_EN

File: 6v6.kicad_sch

2V8

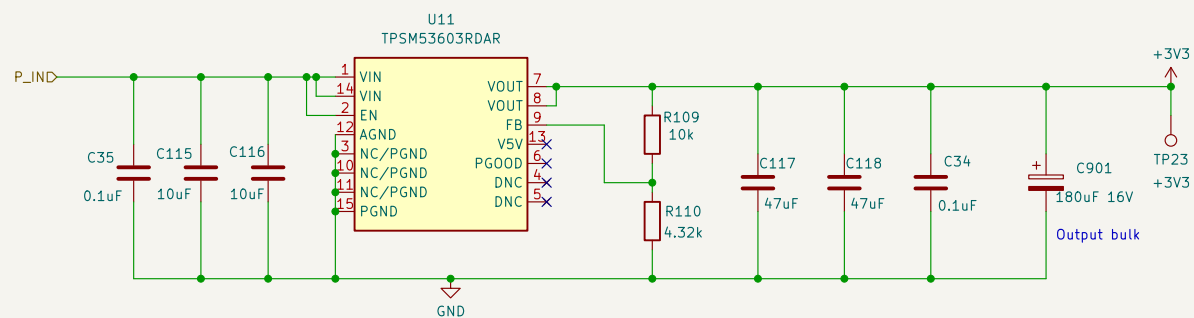
2V8_EN

File: 2v8.kicad_sch

- The 12 V converter is provisionally designed for 6 A continuous and actuator inrush transients.
- Check the U10 RPM0026A and Q100/Q101 TO-263-3 land patterns before PCB placement.
- Input fuse/TVS and reverse-polarity protection remain provisional until the source is frozen.

Sheet: /Power/ File: Power.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 3/24

REGULATED +3V3 / 3 A



Sheet: /Power/3V3/
File: 3v3.kicad_sch

Title:

Size: A4

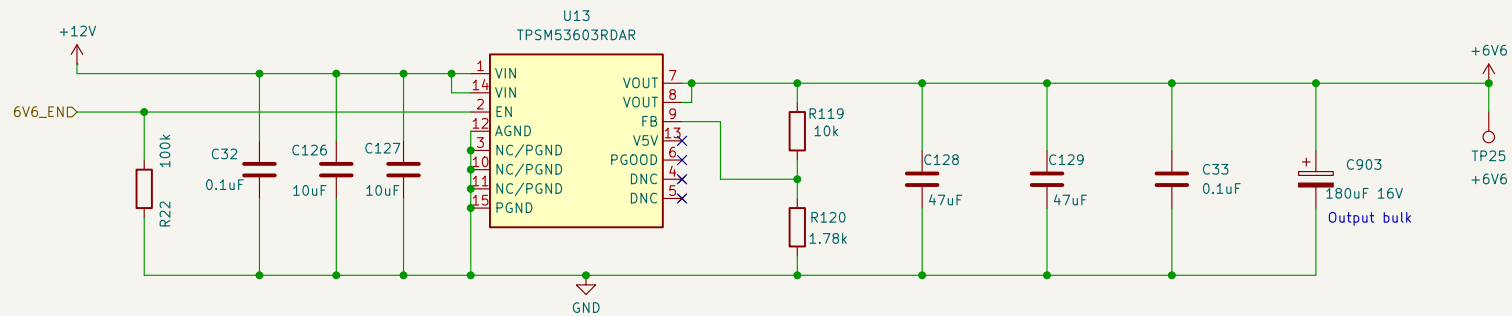
Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 4/24

REGULATED +6V6 / 3 A



Sheet: /Power/6V6/
File: 6v6.kicad_sch

Title:

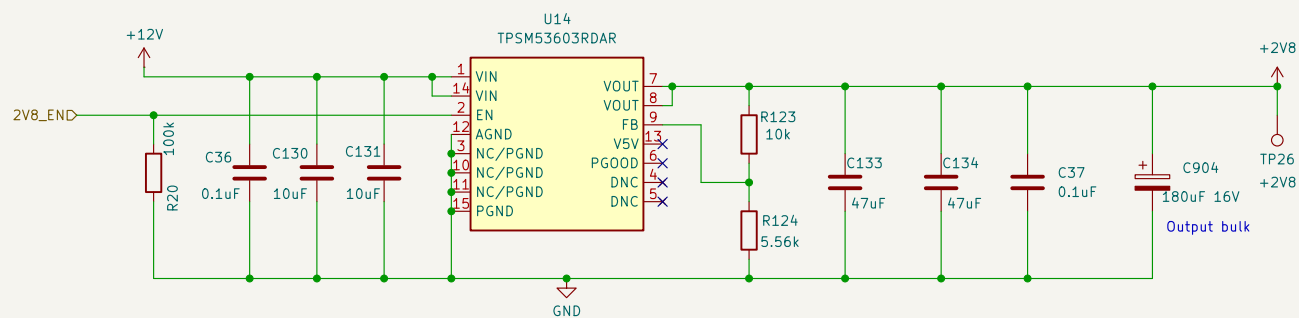
Size: A4 Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 5/24

REGULATED +2V8 / 3 A



Sheet: /Power/2V8/
File: 2v8.kicad_sch

Title:

Size: A4

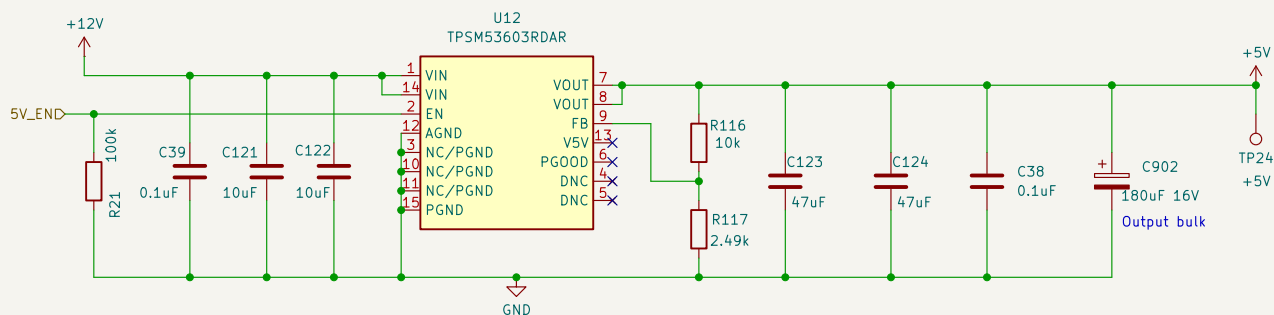
Date:

KiCad E.D.A. 10.0.4

Rev:

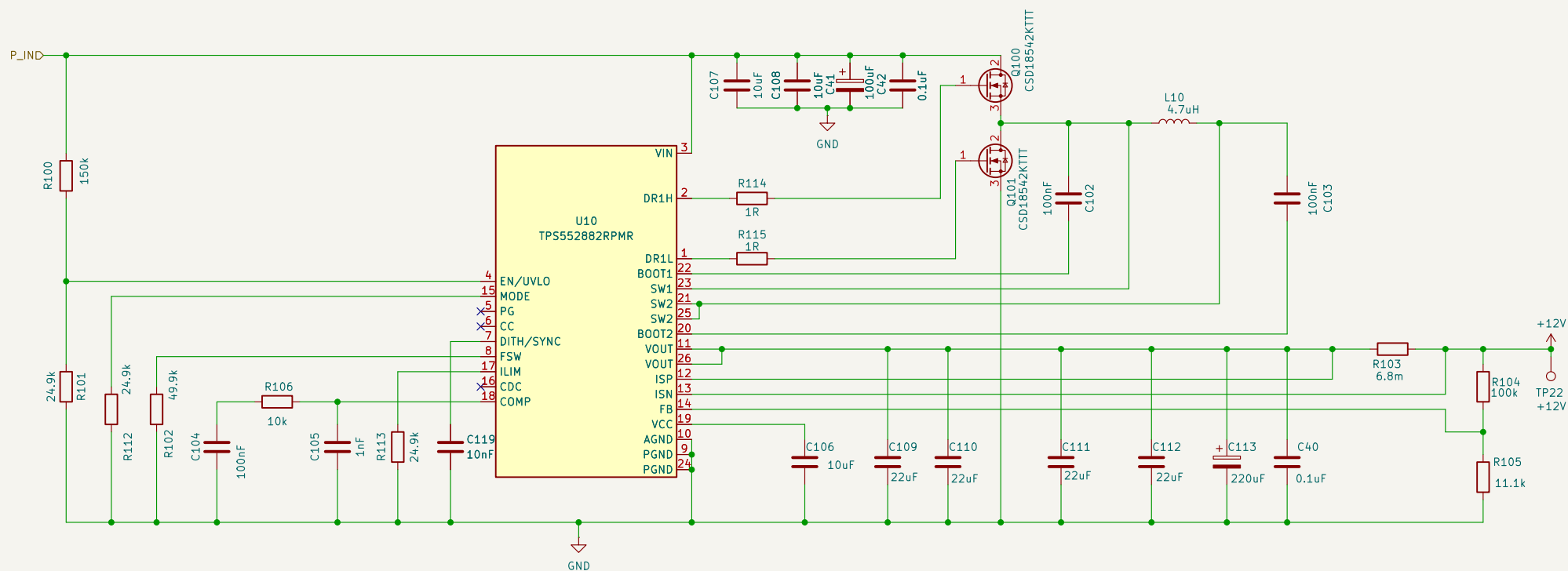
Id: 6/24

REGULATED +5V / 3 A

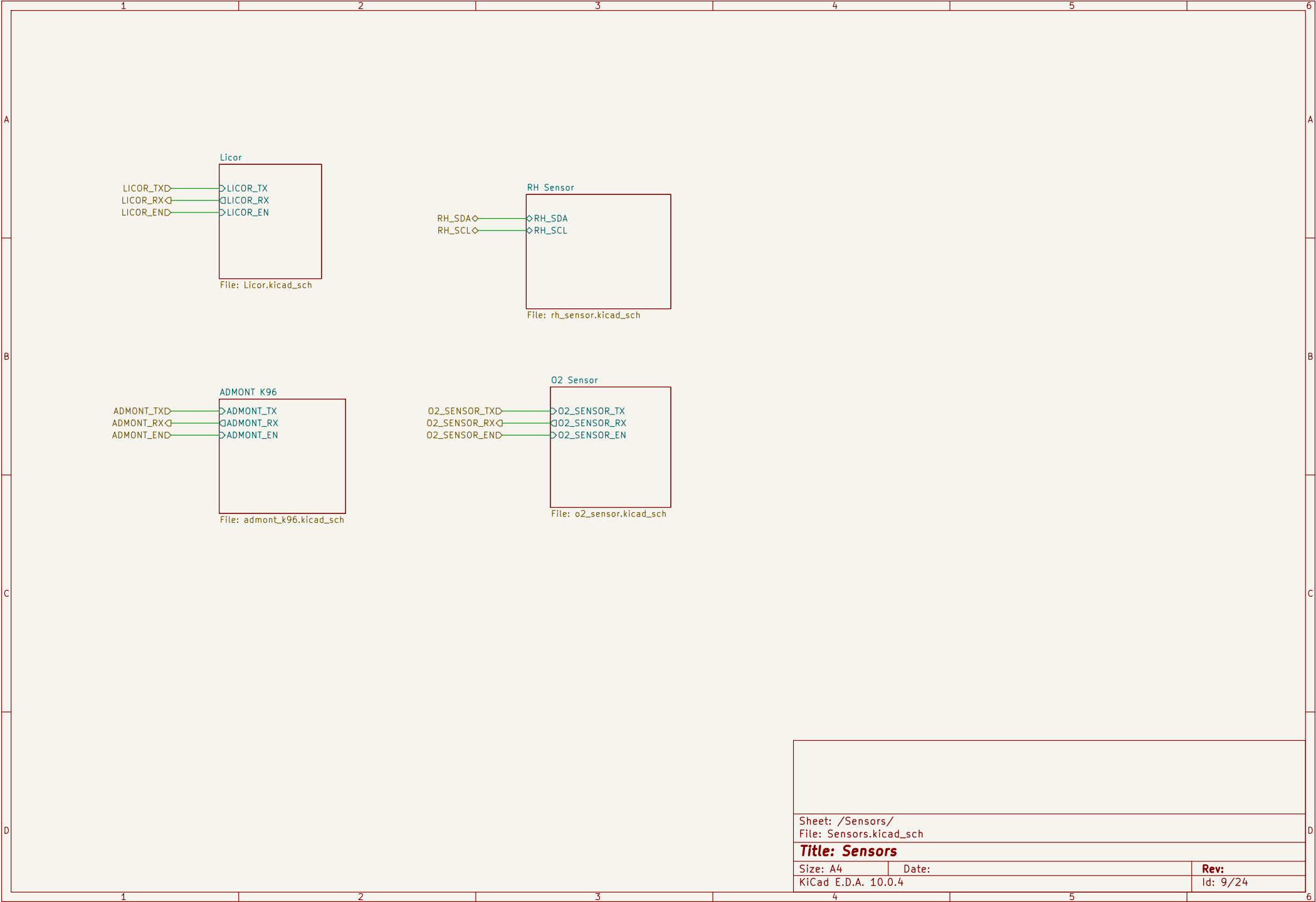


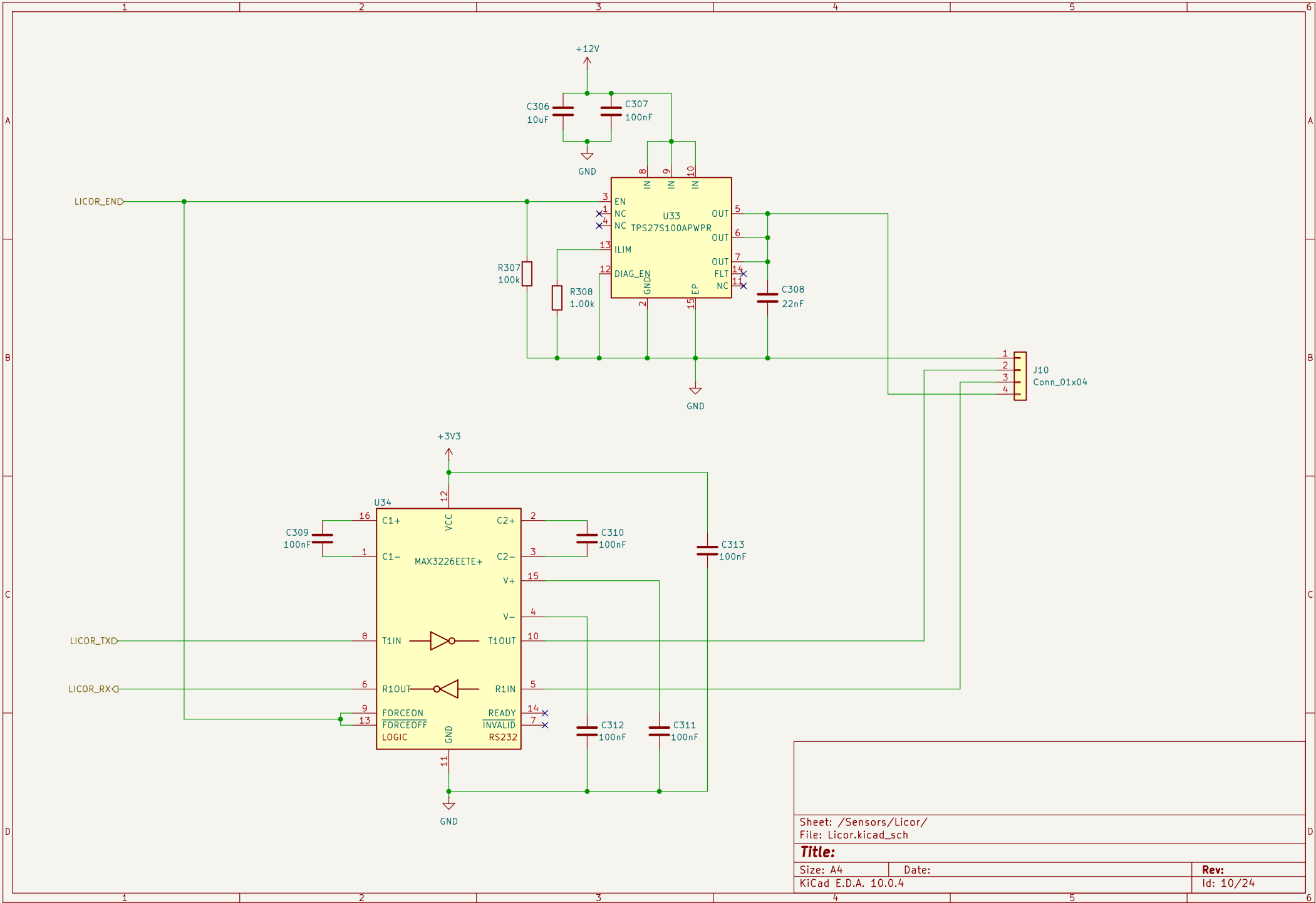
Sheet: /Power/5V/ File: 5v.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 7/24

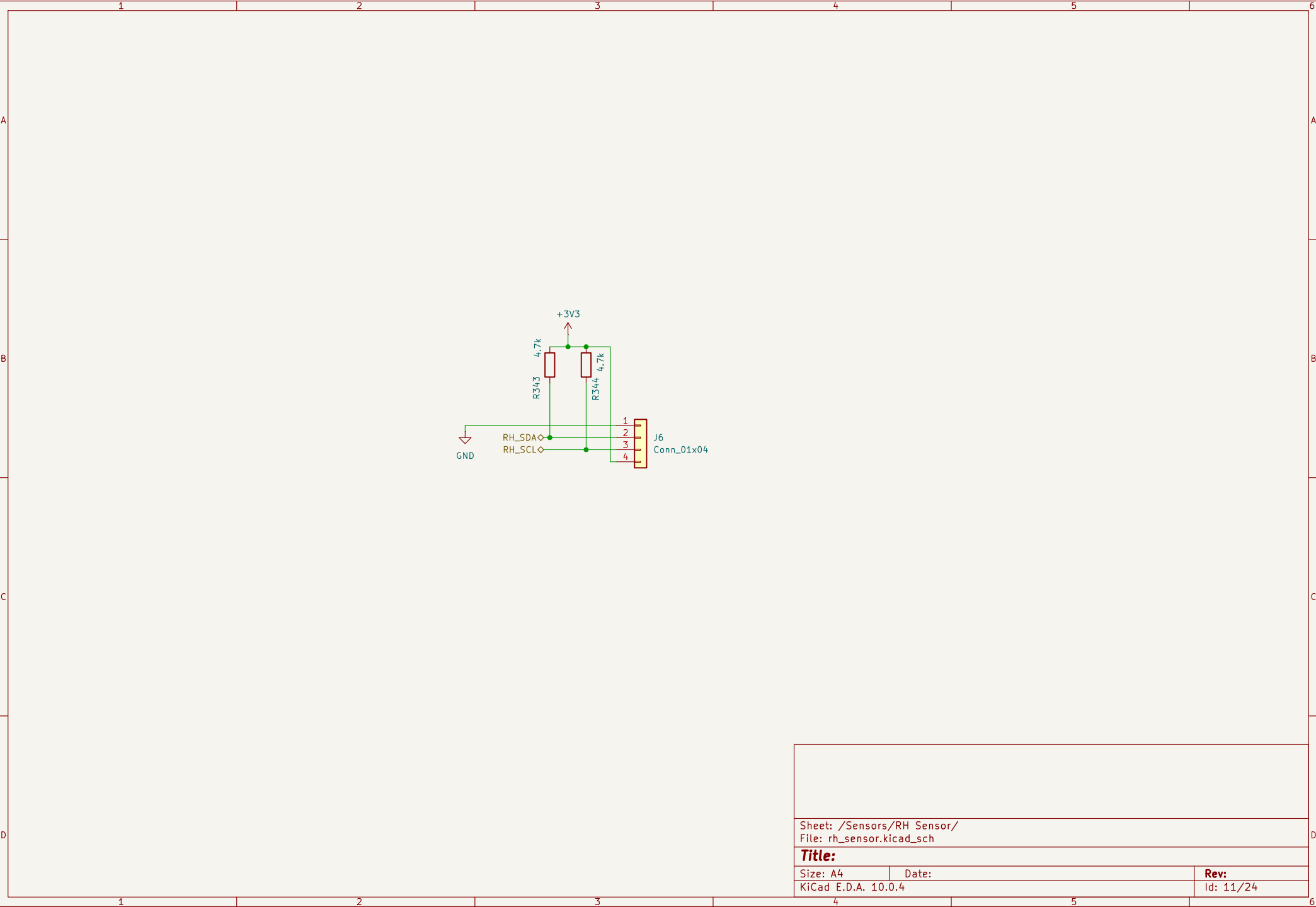
TPS552882, 400 kHz, PFM current limit approximately 7.3 A



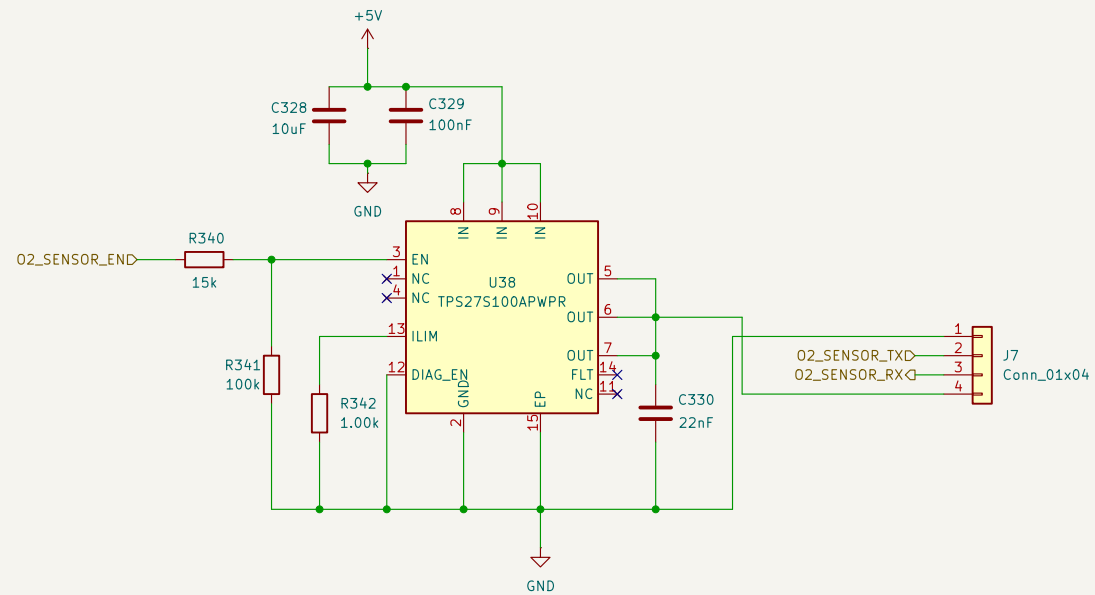
Id: 8/24







Sheet: /Sensors/RH Sensor/ File: rh_sensor.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 11/24



Sheet: /Sensors/O2 Sensor/
File: o2_sensor.kicad_sch

Title:

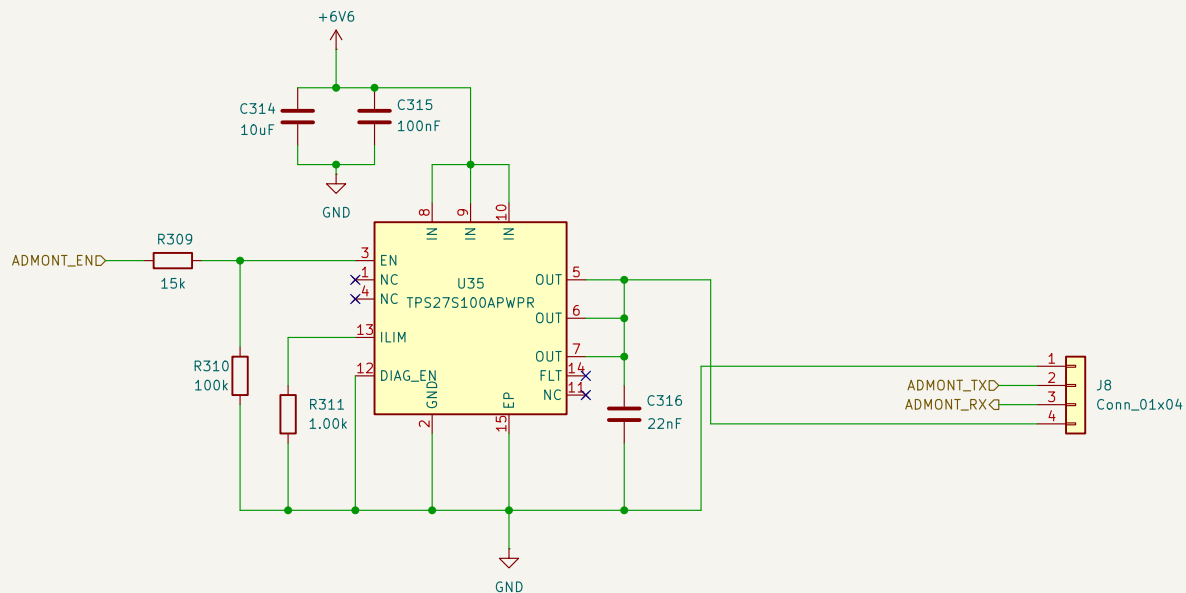
Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 12/24



Sheet: /Sensors/ADMONT K96/
File: admont_k96.kicad_sch

Title:

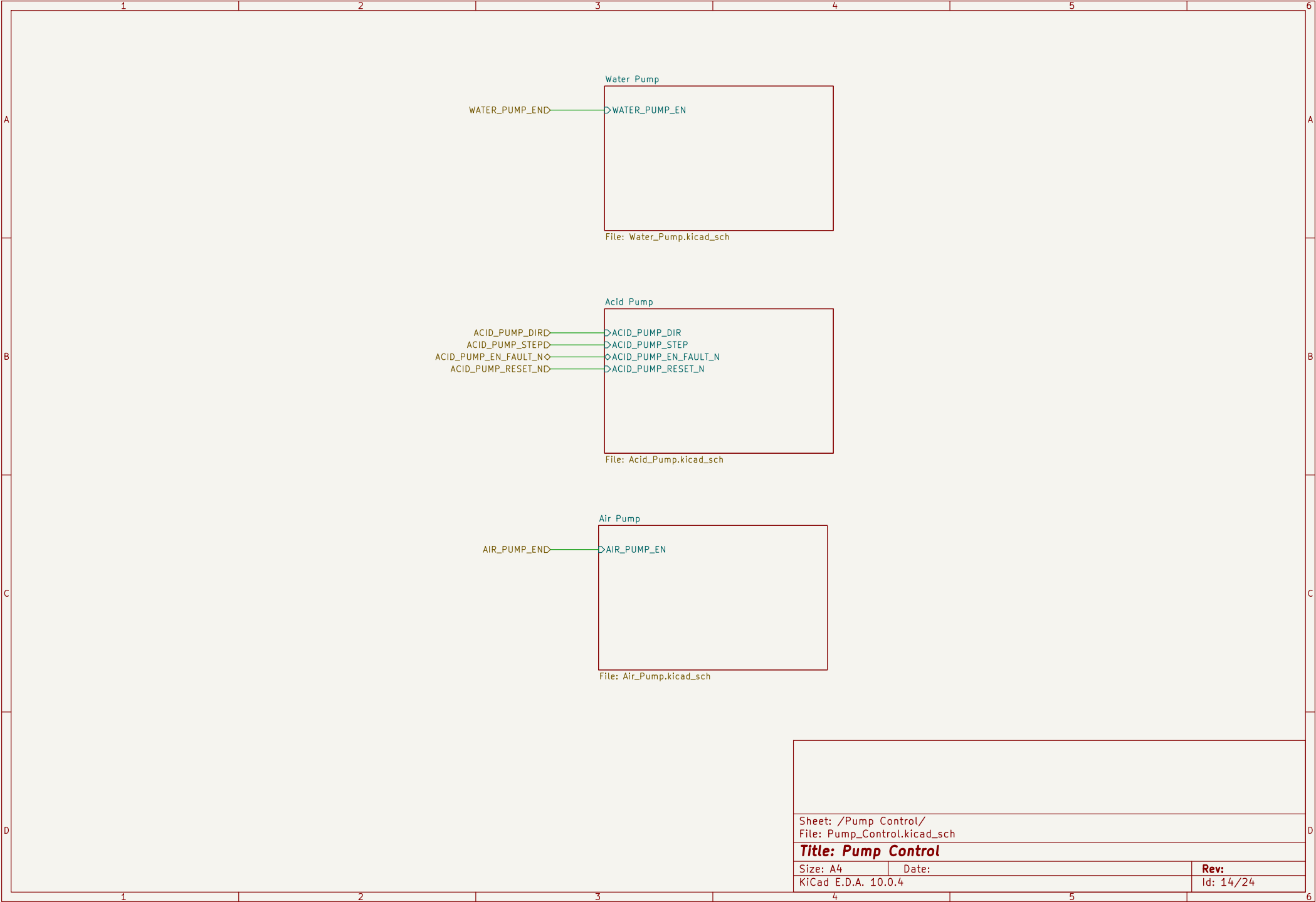
Size: A4

Date:

Rev:

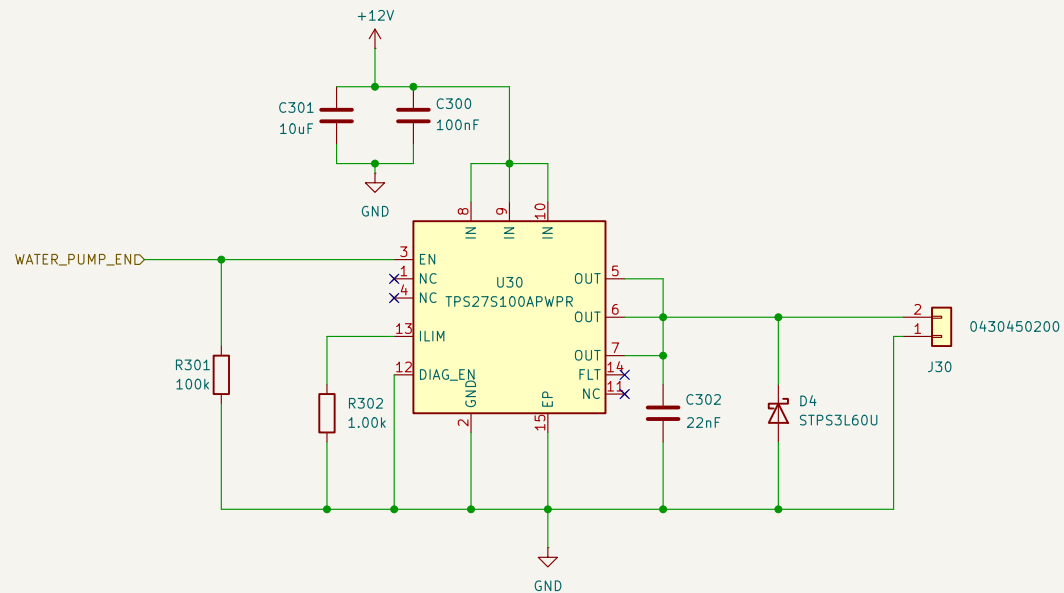
KiCad E.D.A. 10.0.4

Id: 13/24



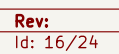
Sheet: /Pump Control/ File: Pump_Control.kicad_sch		
Title: Pump Control		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 14/24

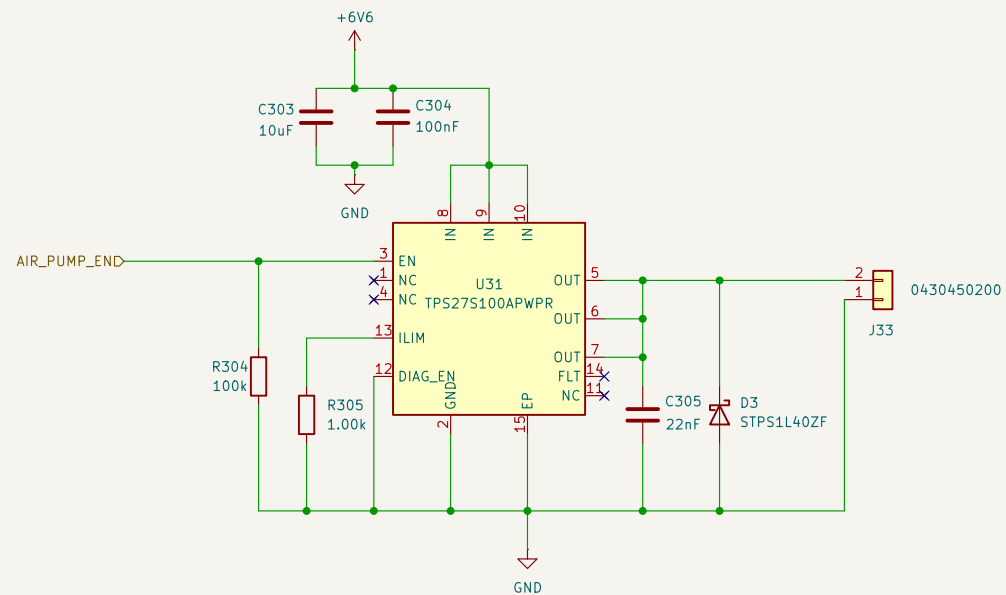
Pump has internal reverse-polarity diode; D4 and TPS27S100A provide turn-off clamping.



Id: 15/24

Design basis: ~0.11 A provides margin over the Aquatech BD202411-T 0.1 A output. Validate the installed RP-TX winding and ramp STEP rate.





Sheet: /Pump Control/Air Pump/
File: Air_Pump.kicad_sch

Title: Air Pump

Size: A4

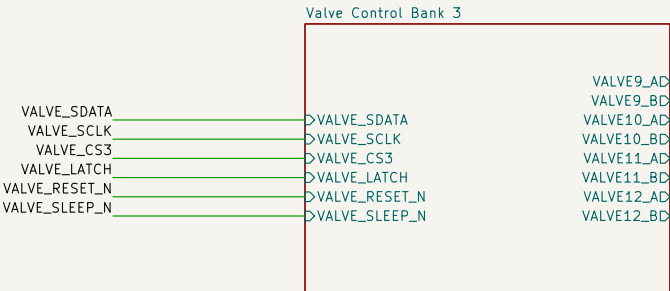
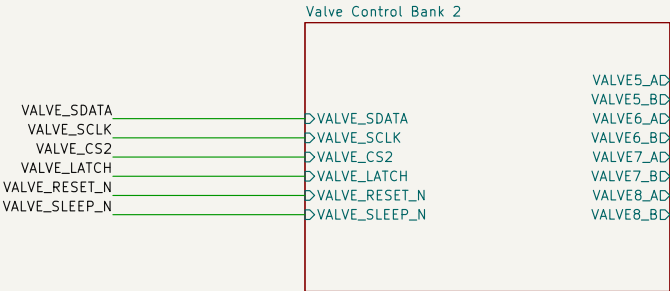
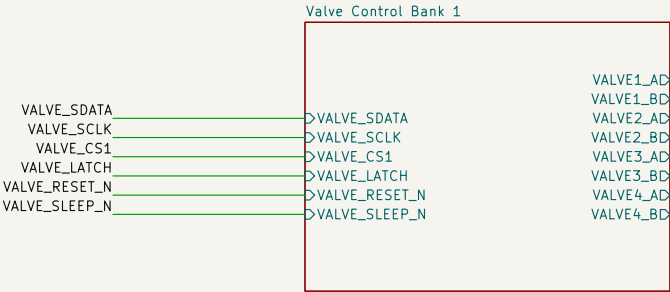
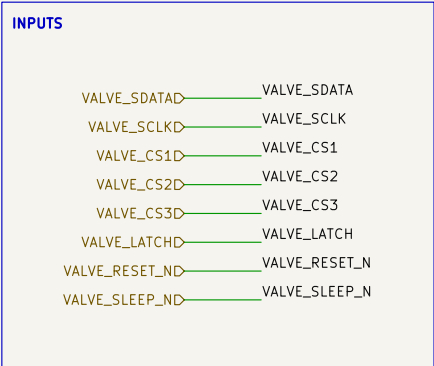
Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 17/24

VALVE CONTROL – THREE CHILD BANKS, 12 POLARITY-REVERSING CHANNELS



Channels 11–12 are reserved spares in the 10–valve population
Three DRV8823–Q1 child banks; 12 H–bridges for 4/8/10–valve variants

Sheet: /Valve Control/
File: Valve_Control.kicad_sch

Title: Valve Control

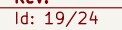
Size: A4

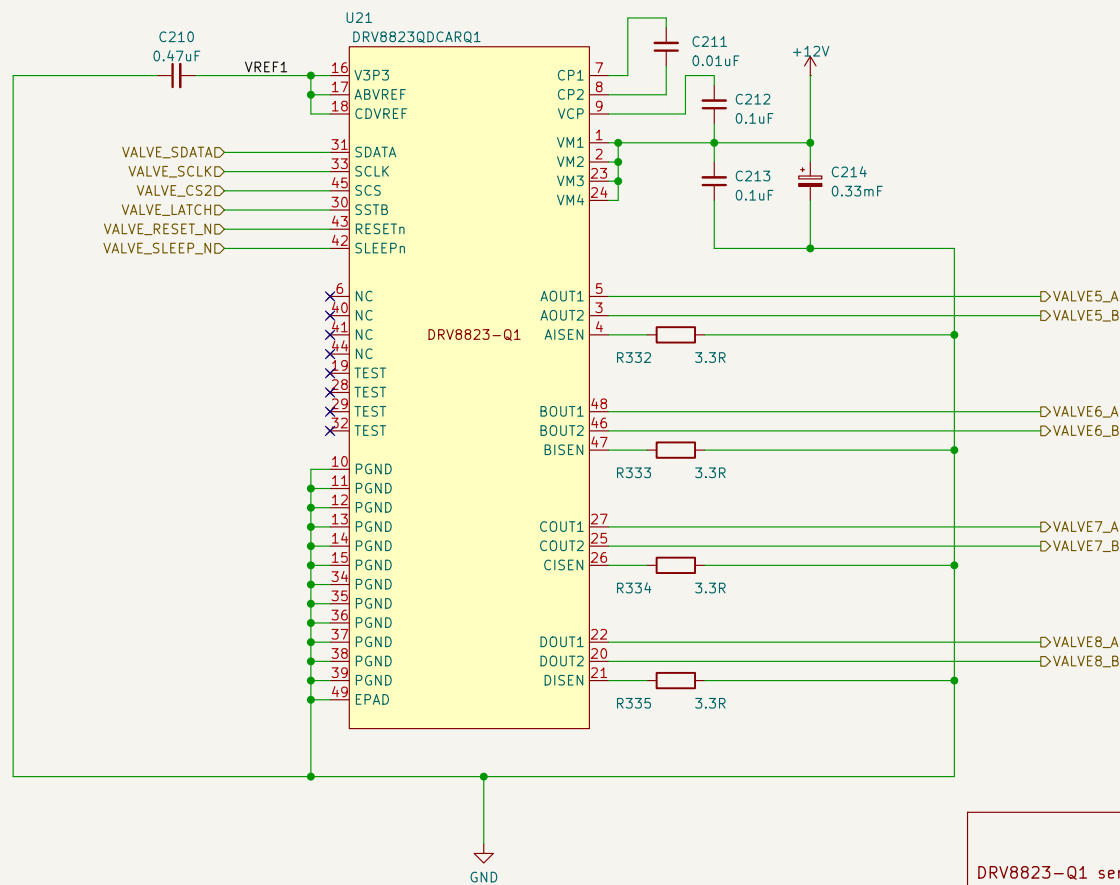
Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 18/24





DRV8823-Q1 serial quad H-bridge
Valves 5-8; fit for 8/10-valve variants

Sheet: /Valve Control/Valve Control Bank 2/
File: Valve_Control_Bank2.kicad_sch

Title: Valve Control Bank 2

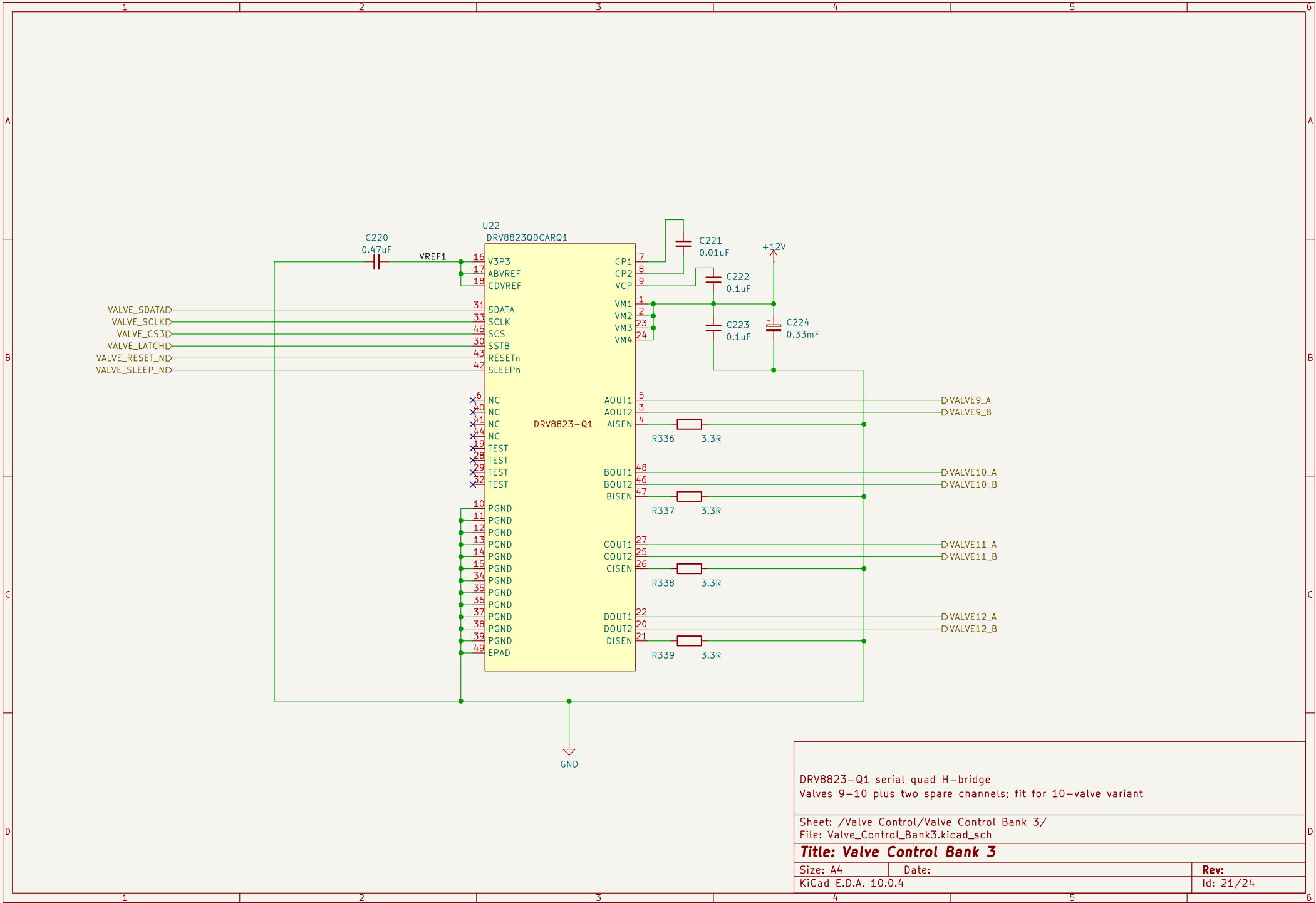
Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.4

Id: 20/24

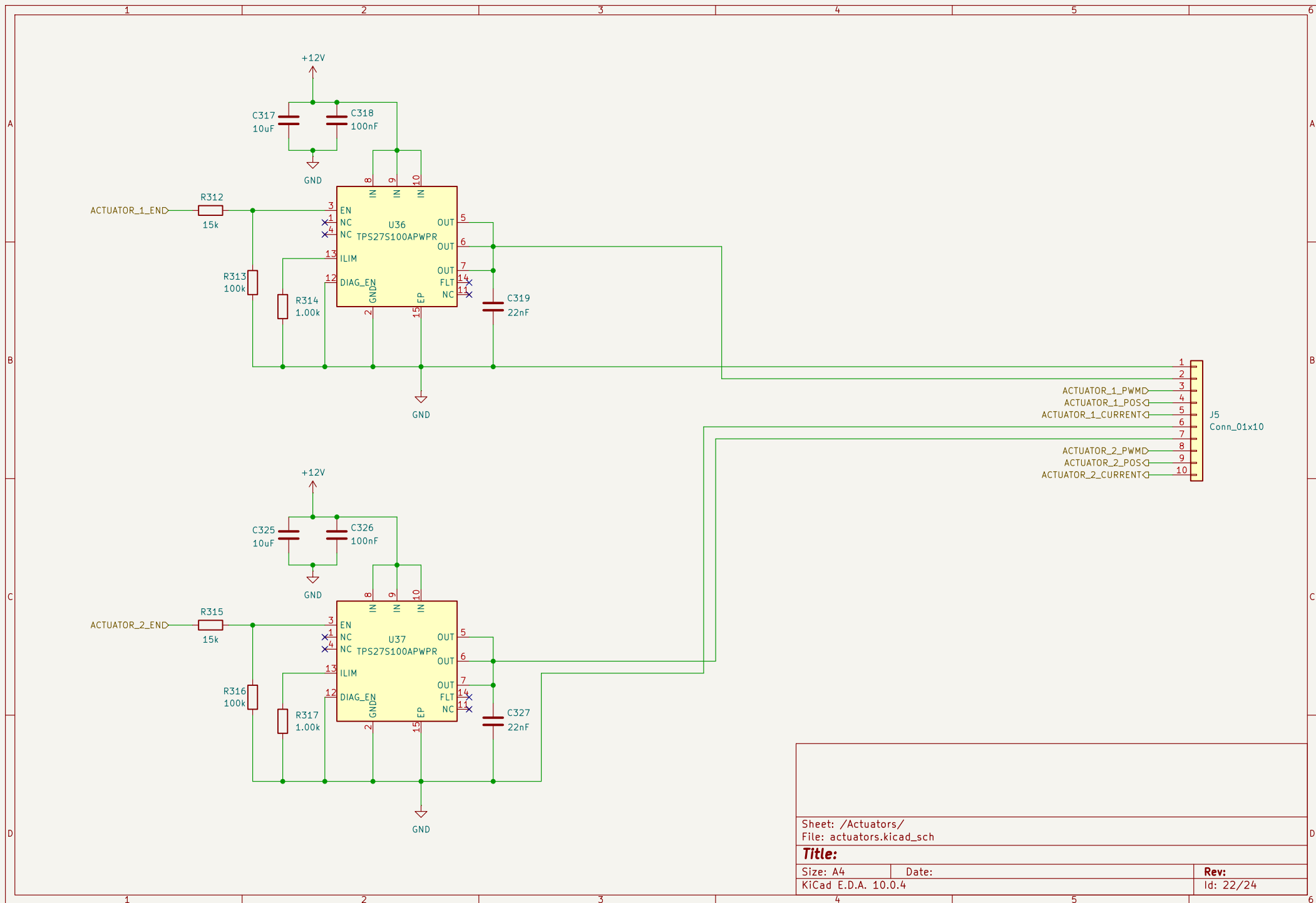


DRV8823-Q1 serial quad H-bridge
Valves 9-10 plus two spare channels; fit for 10-valve variant

Sheet: /Valve Control/Valve Control Bank 3/
File: Valve_Control_Bank3.kicad_sch

Title: Valve Control Bank 3

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 21/24



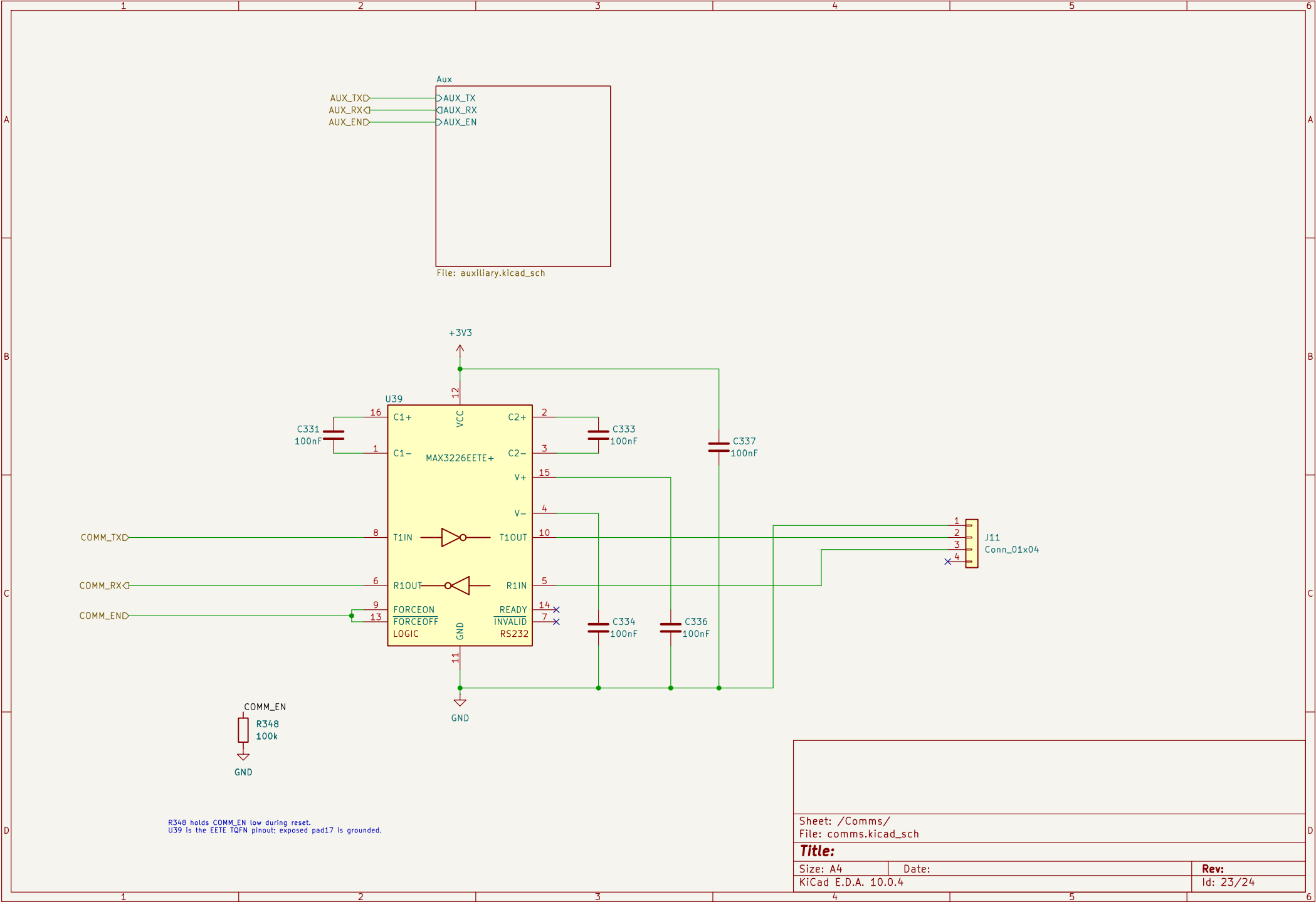
Sheet: /Actuators/
File: actuators.kicad_sch

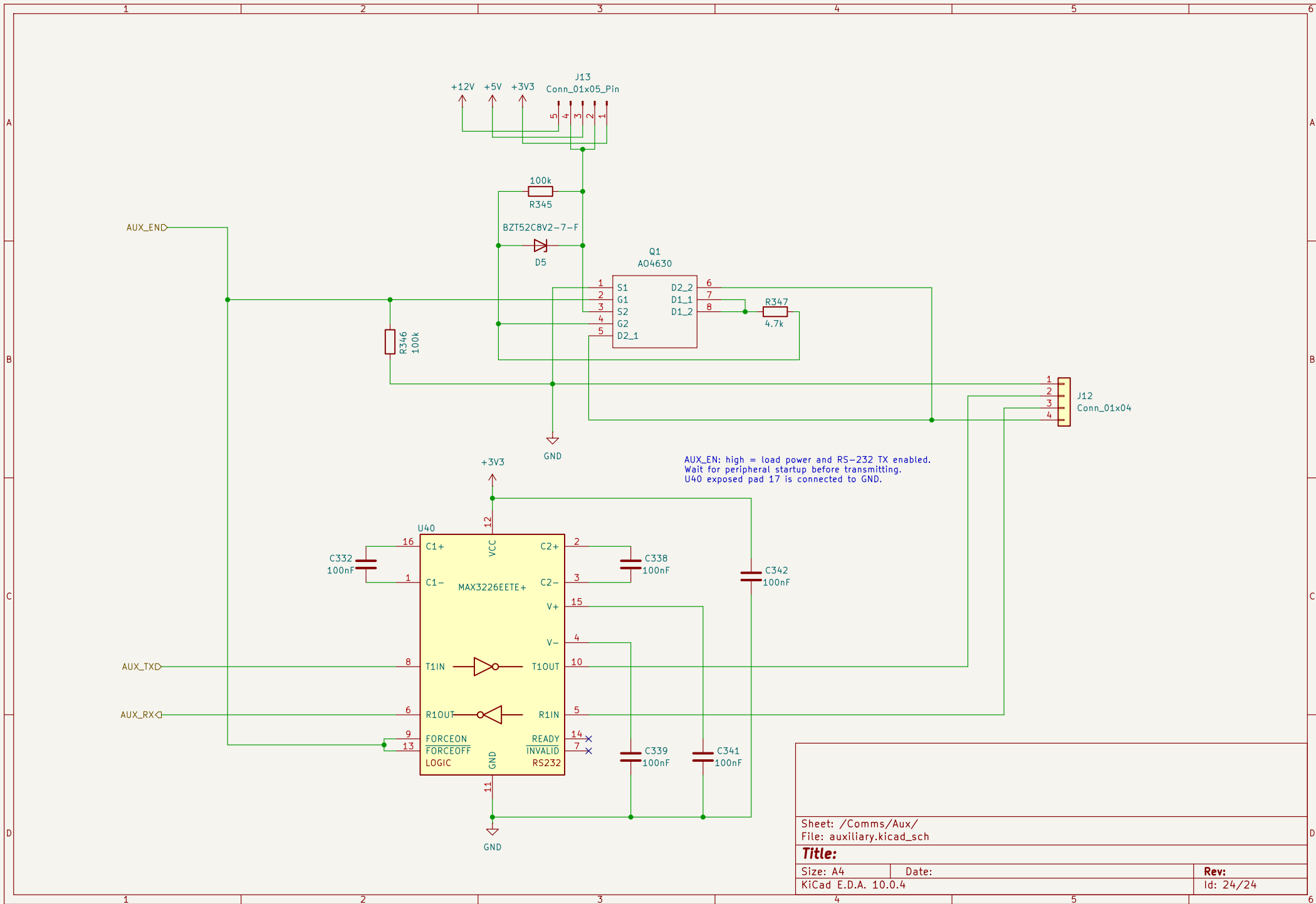
Title:

Size: A4
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 22/24





Sheet: /Comms/Aux/
File: auxiliary.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 24/24